

RAG EVALUATION METRICS CHEATSHEET

AI Engineer Ready · Week 6 — RAG Optimization + Evaluation + LangSmith

Golden Rule -> faithfulness = no hallucination, relevance = answers the question -- check both, not just one

1. Why Evaluation Actually Matters

Reason	The Problem It Solves
Scale	You can't personally read ten thousand answers before every release.
Regression	Did last night's prompt tweak help or quietly hurt? Only a metric tells you.
Trust	Stakeholders want numbers before they'll let you ship to real users.

An answer can feel right and still be wrong. Evaluation isn't extra work on top of a RAG project -- once you're past the demo stage, it IS the project.

2. Faithfulness vs. Relevance

Metric	The Question It Answers
Faithfulness	Is the answer actually supported by the retrieved context, or did the model add something that isn't there?
Relevance	Does the answer actually address what was asked, even if every word in it happens to be true?

Faithfulness = no hallucination. Relevance = actually answers the question. Classic trap: ask "What is the capital of France?" and get back "Paris has a population of about 2 million." That answer is true and fully faithful to real facts -- and completely irrelevant to what was asked.

3. Reading the Two Scores Together

Faithfulness	Relevance	What It Means
High	High	The answer you want: grounded and on-topic.
High	Low	Correct facts, wrong question answered.
Low	High	On-topic, but the model made up the details -- a hallucination that sounds relevant.
Low	Low	Both retrieval and generation failed -- start by checking what was retrieved.

A high faithfulness score with a low relevance score usually means one thing: correct answer, wrong question. Always look at both numbers, never just one.

4. How a Score Gets Produced

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$ Question + Context + Answer
$ -> Faithfulness score (0-1): does the answer follow from the context?
$ -> Relevance score (0-1): does the answer address the question?
$ -> Combined score: both checked together before you trust the result
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In practice, both scores are usually produced by asking a second LLM to compare the answer against the context (for faithfulness) and against the original question (for relevance) -- this is the same idea behind LLM-as-a-Judge, covered next.

Pitfall

Never ship on a single averaged number. A pipeline that averages faithfulness and relevance into one score can look "fine" at 0.75 while hiding a system that is either always off-topic or always making things up -- you need both numbers reported separately to diagnose which.

Remember

You can't fix what you can't measure. Faithfulness and Relevance are the two questions every RAG answer needs to survive, and they fail in different ways -- so they need to be tracked separately.

INTERVIEW PREP

Q: What's the difference between faithfulness and relevance in RAG evaluation?

A: Faithfulness checks whether the answer is grounded in the retrieved context (no hallucination). Relevance checks whether the answer actually addresses the user's question, independent of whether it's true.

Q: Can an answer be faithful but still bad?

A: Yes -- if it's 100% supported by the retrieved context but doesn't answer what was asked, it's faithful and irrelevant, which is still a failed answer from the user's point of view.

Q: How would you evaluate a RAG system at scale without reading every answer?

A: Run a scoring pipeline (hand-rolled prompts or a library like RAGAS) that computes faithfulness and relevance for every question-context-answer triple, then track the aggregate and the distribution, not just one average number.